Constructions With Monoidal Categories

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01UF This chapter contains some material on constructions with monoidal categories.

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1 Moduli Categories of Monoidal Structures

1.1 The Moduli Category of Monoidal Structures on a Category

Let $\mathcal{C}$ be a category.

01UJ **Definition 1.1.1.1.** The **moduli category of monoidal structures on** $\mathcal{C}$ is the category $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ defined by

$$\mathcal{M}_{\mathbb{E}_1}(\mathcal{C}) \overset{\text{def}}{=} \mathsf{pt} \times_{\mathsf{Cats}} \mathsf{MonCats}, \qquad \begin{tikzcd} \mathcal{M}_{\mathbb{E}_1}(\mathcal{C}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathsf{MonCats} \arrow[d, "\text{忘}"] \\ \mathsf{pt} \arrow[r, "{[\mathcal{C}]}"'] & \mathsf{Cats}. \end{tikzcd}$$

01UK **Remark 1.1.1.2.** In detail, **the moduli category of monoidal structures on** $\mathcal{C}$ is the category $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ where:

- *Objects.* The objects of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ are monoidal categories $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ whose underlying category is $\mathcal{C}$.

- *Morphisms.* A morphism from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ is a strong monoidal functor structure

$$\begin{aligned} \mathrm{id}^{\otimes}_{\mathcal{C}} &\colon A \boxtimes_{\mathcal{C}} B \overset{\sim}{\dashrightarrow} A \otimes_{\mathcal{C}} B, \\ \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}} &\colon \mathbb{1}'_{\mathcal{C}} \overset{\sim}{\dashrightarrow} \mathbb{1}_{\mathcal{C}} \end{aligned}$$

  on the identity functor $\mathrm{id}_{\mathcal{C}} \colon \mathcal{C} \to \mathcal{C}$ of $\mathcal{C}$.

- *Identities.* For each $M \overset{\text{def}}{=} \left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right) \in \mathrm{Obj}(\mathcal{M}_{\mathbb{E}_1}(\mathcal{C}))$, the unit map

$$\mathbb{1}^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,M} \colon \mathrm{pt} \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, M)$$

  of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ at M is defined by

$$\mathrm{id}^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M} \overset{\text{def}}{=} \left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right),$$

  where $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ is the identity monoidal functor of $\mathcal{C}$ of **??**.

- *Composition.* For each $M, N, P \in \mathrm{Obj}(\mathcal{M}_{\mathbb{E}_1}(\mathcal{C}))$, the composition map

$$\circ^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,N,P} \colon \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(N, P) \times \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, N) \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, P)$$

  of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ at (M, N, P) is defined by

$$\left(\mathrm{id}^{\otimes,\prime}_{\mathcal{C}}, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|\mathcal{C}}\right) \circ^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,N,P} \left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right) \overset{\text{def}}{=} \left(\mathrm{id}^{\otimes,\prime}_{\mathcal{C}} \circ \mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|\mathcal{C}} \circ \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right).$$

01UL **Remark 1.1.1.3.** In particular, a morphism in $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ satisfies the following conditions:

01UM 1. *Naturality.* For each pair $f\colon A \to X$ and $g\colon B \to Y$ of morphisms of $\mathcal{C}$, the diagram

$$\begin{tikzcd}
A \boxtimes_{\mathcal{C}} B \arrow[r, "f \boxtimes_{\mathcal{C}} g"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B}"'] & X \boxtimes_{\mathcal{C}} Y \arrow[d, "\mathrm{id}^{\otimes}_{X,Y}"] \\
A \otimes_{\mathcal{C}} B \arrow[r, "f \otimes_{\mathcal{C}} g"'] & X \otimes_{\mathcal{C}} Y
\end{tikzcd}$$

commutes.

01UN 2. *Monoidality.* For each $A, B, C \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] & & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"'] & & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
 & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

commutes.

01UP 3. *Left Monoidal Unity.* For each $A \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} A \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}'_{\mathcal{C}}, A}"] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} A \arrow[dr, "\lambda^{\mathcal{C}}_A"] & \\
\mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} A \arrow[ur, "\mathrm{id}^{\otimes}_{\mathbb{1}} \boxtimes_{\mathcal{C}} \mathrm{id}_A"] \arrow[rrr, "\lambda^{\mathcal{C},\prime}_A"'] & & & A
\end{tikzcd}$$

commutes.

01UQ 4. *Right Monoidal Unity.* For each $A \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & A \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{A,\mathbb{1}'_{\mathcal{C}}}"] & A \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[dr, "\rho^{\mathcal{C}}_{A}"] & \\
A \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[ur, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[rrr, "\rho^{\mathcal{C},\prime}_{A}"'] & & & A
\end{tikzcd}$$

commutes.

01UR **Proposition 1.1.1.4.** Let $\mathcal{C}$ be a category.

01US 1. *Extra Monoidality Conditions.* Let $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ be a morphism of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$.

01UT (a) The diagram

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"'] & (A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"] \\
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] & (A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C
\end{tikzcd}$$

commutes.

01UU (b) The diagram

$$\begin{tikzcd}
A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A, B \boxtimes_{\mathcal{C}} C}"'] & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

commutes.

0288 2. *Extra Monoidal Unity Constraints.* Let $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ be a morphism of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$.

0289 (a) The diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

commutes.

028A (b) The diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

commutes.

028B (c) The diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"'] & \mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}
\end{tikzcd}$$

commutes.

028C (d) The diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"'] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}
\end{tikzcd}$$

commutes.

01UV 3. *Mixed Associators.* Let $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ and $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ be monoidal structures on $\mathcal{C}$ and let

$$\mathrm{id}^{\otimes}_{-_1,-_2}\colon -_1 \boxtimes_{\mathcal{C}} -_2 \to -_1 \otimes_{\mathcal{C}} -_2$$

be a natural transformation.

01UW (a) If there exists a natural transformation

$$\alpha^{\otimes}_{A,B,C}\colon (A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \to A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A\otimes_{\mathcal{C}} B,C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C}}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_{A,B\boxtimes_{\mathcal{C}} C}"] \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\otimes}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 1.1.1.3.

01UX (b) If there exists a natural transformation

$$\alpha^{\boxtimes}_{A,B,C}\colon (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \to A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)$$

making the diagrams

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A,B\otimes_{\mathcal{C}} C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C}}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"'] & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 1.1.1.3.

01UY (c) If there exists a natural transformation

$$\alpha^{\boxtimes,\otimes}_{A,B,C} \colon (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \to A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)$$

making the diagrams

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C}}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\alpha^{\mathcal{C},\prime}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A, B \boxtimes_{\mathcal{C}} C}"] \\
(A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 1.1.1.3.

Proof. *Item 1, Extra Monoidality Conditions*: We claim that Items 1a and 1b are indeed true:

1. *Proof of Item 1a:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms $\mathrm{id}^{\otimes}_{A,B}$ and id_C.

2. *Proof of Item 1b:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms id_A and $\mathrm{id}^{\otimes}_{B,C}$.

This finishes the proof.
Item 2, Extra Monoidal Unity Constraints: We claim that Items 2a and 2b are indeed true:

1. *Proof of Item 1a:* Indeed, consider the diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[rr, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[dr, "\mathrm{id}_{\mathbb{1}_{\mathcal{C}}} \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[dd, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] & (1) & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\mathrm{id}_{\mathbb{1}_{\mathcal{C}}} \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[dddd, bend left, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[ddr, equal] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[drr, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & (3) & (2) \\
& & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}"] \quad (4) \\
& & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

whose boundary diagram is the diagram whose commutativity we wish to prove. Since:

- Subdiagram (1) commutes by the naturality of $\mathrm{id}_{\mathcal{C}}^{\otimes,-1}$;
- Subdiagram (2) commutes trivially;
- Subdiagram (3) commutes by the naturality of $\lambda^{\mathcal{C}}$, where the equality $\rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}$ comes from **??**;
- Subdiagram (4) commutes by the right monoidal unity of $\left(\mathrm{id}_{\mathcal{C}}, \mathrm{id}_{\mathcal{C}}^{\otimes}, \mathrm{id}_{\mathcal{C}|\mathbb{1}}^{\otimes}\right)$;

so does the boundary diagram, and we are done.

2. *Proof of Item 1b:* Indeed, consider the diagram

$$
\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[rr, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}'_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, "\mathrm{id}^{\otimes}_{\mathbb{1}} \otimes_{\mathcal{C}} \mathrm{id}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dd, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] \arrow[drr, phantom, "(1)"] & & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\mathrm{id}^{\otimes}_{\mathbb{1}} \boxtimes_{\mathcal{C}} \mathrm{id}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[ddd, bend left, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, equal] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}_{\mathcal{C}}}"] \arrow[dl, phantom, "(2)"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[ddr, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] \arrow[rr, phantom, "(3)"] & & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dl, phantom, "(4)"] \\
& & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}
$$

whose boundary diagram is the diagram whose commutativity we wish to prove. Since:

- Subdiagram (1) commutes by the naturality of $\mathrm{id}^{\otimes,-1}_{\mathcal{C}}$;
- Subdiagram (2) commutes trivially;

- Subdiagram (3) commutes by the naturality of $\rho^{\mathcal{C}}$, where the equality $\rho^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}} = \lambda^{\mathcal{C}}_{\mathbb{1}_{\mathcal{C}}}$ comes from **??**;
- Subdiagram (4) commutes by the left monoidal unity of $\left(\mathrm{id}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathcal{C}|\mathbb{1}}\right)$;

so does the boundary diagram, and we are done.

3. *Proof of Item 2c:* Indeed, consider the diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] \arrow[dr, phantom, "(1)"] & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}.
\end{tikzcd}$$

Since:

- The boundary diagram commutes trivially;
- Subdiagram (1) commutes by Item 1b;

it follows that the diagram

$$\begin{tikzcd}
\mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] & \mathbb{1}'_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"'] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}'_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[d, "\rho^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}'_{\mathcal{C}}
\end{tikzcd}$$

commutes. But since $\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}$ is an isomorphism, it follows that the diagram (†) also commutes, and we are done.

4. *Proof of Item 2d:* Indeed, consider the diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"'] \arrow[dr, phantom, "(1)"] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
\mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

Since:

- The boundary diagram commutes trivially;
- Subdiagram (1) commutes by Item 1a;

it follows that the diagram

$$\begin{tikzcd}
\mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] & \mathbb{1}_{\mathcal{C}} \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}_{\mathcal{C}},\mathbb{1}'_{\mathcal{C}}}"] \arrow[d, "\rho^{\mathcal{C},\prime}_{\mathbb{1}_{\mathcal{C}}}"'] \arrow[dr, phantom, "(\dagger)"] & \mathbb{1}_{\mathcal{C}} \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[d, "\lambda^{\mathcal{C}}_{\mathbb{1}'_{\mathcal{C}}}"] \\
& \mathbb{1}_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes,-1}_{\mathbb{1}}"'] & \mathbb{1}_{\mathcal{C}}
\end{tikzcd}$$

commutes. But since $\mathrm{id}^{\otimes,-1}_{\mathbb{1}}$ is an isomorphism, it follows that the diagram (†) also commutes, and we are done.

This finishes the proof.

Item 3, Mixed Associators: We claim that Items 3a to 3c are indeed true:

01UZ 1. *Proof of Item 3a:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of Item 2 of Definition 1.1.1.3 as follows:

$$\begin{tikzcd}
& (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dd, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] \arrow[dr, phantom, "(1)"] & (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\otimes}_{A,B,C}"] \arrow[d, phantom, "(2)"] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[dd, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"'] & (3) & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
& A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

Since:

- Subdiagram (1) commutes by Item 1a of Item 1.

- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by assumption.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 1.1.1.3.

01V0 2. *Proof of Item 3b:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of Item 2 of Definition 1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dr, dashed, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[dd, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] & (1) & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[dl, dashed, "\mathrm{id}^{\otimes}_{A, B \boxtimes_{\mathcal{C}} C}"'] \arrow[dd, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
 & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[dd, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"'] & (2) \quad (3) & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
 & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

Since:

- Subdiagram (1) commutes by assumption.
- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by Item 1b of Item 1.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 1.1.1.3.

01V1 3. *Proof of Item 3c:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$

of Item 2 of Definition 1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] \arrow[dr, "\alpha^{\mathcal{C}}_{A,B,C}"] & \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[dd, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] \arrow[ddr, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[r, phantom, "(1)"] & & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[dd, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \\
 & & \\
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[dr, "\alpha^{\mathcal{C},\prime}_{A,B,C}"'] \arrow[r, phantom, "(2)"] & & A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"] \\
 & A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) &
\end{tikzcd}$$

Since subdiagrams (1) and (2) commute by assumption, it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Definition 1.1.1.3.

This finishes the proof. $\square$

1.2 The Moduli Category of Braided Monoidal Structures on a Category

1.3 The Moduli Category of Symmetric Monoidal Structures on a Category

2 Moduli Categories of Closed Monoidal Structures

3 Moduli Categories of Refinements of Monoidal Structures

3.1 The Moduli Category of Braided Refinements of a Monoidal Structure

Appendices

A Other Chapters

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1. Sets
2. Constructions With Sets
3. Monoidal Structures on the Category of Sets
4. Pointed Sets
5. Tensor Products of Pointed Sets

Relations

6. Relations
7. Constructions With Relations
8. Equivalence Relations and Apartness Relations

Category Theory

9. Categories

Monoidal Categories

10. Constructions With Monoidal Categories

Bicategories

11. Types of Morphisms in Bicategories